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Test anxiety, psychological adaptability, and learning performance: Effects of a mindfulness training intervention on university students

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Abstract:

BACKGROUND: Within the demanding landscape of higher education, the intersection of academic excellence and students' psychological well-being is a problem. Mindfulness training has emerged as a potential solution for fostering brain connectivity and emotional regulation. However, its influence on test anxiety, psychological adaptability, and academic performance, along with their interrelations, remains insufficiently investigated. To address this void, the present study aimed to examine the impact of a mindfulness training program on university students' test anxiety, adaptability, and learning success and the interrelations among these variables.

METHODS AND MATERIAL: A quasi-experimental design involving 105 university students assigned non-randomly to either a mindfulness training group or a control group was employed. The intervention comprised 12 weekly sessions that integrated theory and activities. Before the intervention and following its completion, test anxiety, and adaptability were measured using standardized scales, whereas individual grade point averages were obtained from the university records. Data were subjected to descriptive analysis, multivariate analysis of covariance, repeated measures analyses of variance, and structural equation modeling.

RESULTS: The test anxiety scores were significantly lower in the experimental group than in the comparison group. Additionally, compared with their nontreated counterparts, the mindfulness group reported a significant increase in adaptability. However, contrary to expectations, the intervention did not yield a significant improvement in academic grades, and no significant direct link between test anxiety and learning performance was established. Furthermore, no mediating role of adaptability between test anxiety and academic performance was found.

CONCLUSIONS: This evidence highlights the potential of mindfulness practices for alleviating test anxiety and improving adaptability. However, the findings highlight the need for further interventions to enhance learning outcomes and explore diverse causal models.

Keywords:

Academic achievement, educational psychology, mindfulness, test anxiety

Introduction

In the bustling sphere of higher education, the quest for academic excellence intertwines with the intricate tapestry

of students' psychological well-being and flexibility, particularly given today's hypercompetitive social environment and issues such as nomophobia.^[1,2] In this sense, the triad of learning performance, psychological adaptability, and academic

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stress, including test anxiety, represent pivotal pillars shaping the academic journey of university students. Test anxiety is a situation-specific manifestation of trait anxiety that involves physical arousal, overexcitement, and cognitive interference during or before academic exams and is often accompanied by distressing thoughts about potential failure and its ramifications.^[3] Adaptability refers to an individual's capacity to self-regulate, making adjustments in thinking, actions, and emotions within unfamiliar, irregular circumstances.^[4] Adaptability stands out because of its emphasis on navigating situational novelty and uncertainty, setting it apart from related concepts such as resilience, which primarily address overcoming challenges.^[5] Moreover, academic performance remains a crucial metric reflecting students' knowledge acquisition and competence.

Adaptability, learning outcomes, and test anxiety are crucial variables to evaluate since they have significant implications for students' mental health, educational accomplishments, and future career opportunities. Test anxiety is a prevalent and debilitating phenomenon among university students, affecting a considerable portion of the population.^[6,7] Substantial efforts have been made to alleviate test anxiety and enhance learning outcomes via diverse approaches, such as attentional cognitive bias modification, which is designed to relieve anxiety, including test anxiety,^[8] and working memory training, which is supposed to influence both test anxiety^[9] and academic achievement.^[10] Attentional cognitive bias modification is a technique that aims to modify the attentional patterns of individuals inclined toward negative or threatening stimuli, which can trigger or maintain anxiety.^[11] A meta-analysis^[12] concluded that attentional cognitive bias modification interventions have moderate effects on a variety of symptoms, including anxiety, with heterogeneous findings across studies. Similarly, working memory training is supposed to mitigate anxiety by enabling a person to hold on to certain information and use it to guide decisions and behavior.^[13] A recent systematic review and meta-analysis^[14] reported the small effectiveness of working memory training interventions in reducing anxiety. While some works have documented the positive effects of working memory training on academic performance, the overall evidence is mixed, being drawn primarily from studies.^[15,16]

Mindfulness training, which is increasingly acknowledged for its role in nurturing emotional regulation,^[17] cognitive flexibility,^[18] stress management,^[19] and psychological well-being,^[20] has emerged as a promising avenue within this realm. Simply put, mindfulness practices are mainly aimed at cultivating positive attitudes toward life events and accepting nonjudgmental present-moment awareness, which is anticipated to diminish reactivity

to stressors and equip individuals with effective coping mechanisms.^[21-23] Mindfulness training has been noted to stimulate brain connectivity and engage neural networks that counter emotional arousal systems, leading to a more composed mental state.^[24] In particular, techniques such as slow breathing, commonly integral to mindfulness programs, are believed to encourage dominance of the parasympathetic autonomic system.^[25]

Despite its roots in ancient contemplative practices, the mindfulness concept has garnered considerable attention in contemporary psychological research. Nevertheless, existing studies focus primarily on singular facets of student well-being or cognitive functions, leaving unexplored territory, which is a comprehensive examination of the impact of mindfulness training on affective traits—specifically, test anxiety and adaptability—along with its influence on academic performance, as well as their interplay within educational contexts. The present study seeks to bridge this gap, aligning with theoretical frameworks (outlined below) that motivated us to suggest the potential of mindfulness in weakening test anxiety while bolstering adaptability and thus improving learning outcomes.

In terms of trial evidence, the monitor and acceptance theory explains the beneficial mental and physical health outcomes of mindfulness activities through the synergy of two emotion regulatory mechanisms: attention monitoring skills and experiential acceptance.^[26] This theoretical view conceptualizes monitoring as the self-reported tendency to monitor one's present-moment experiences, which is supposed to intensify affective reactivity, i.e., how well an individual copes with daily stressors,^[27] whereas acceptance is understood by the authors as an orientation of receptivity and noninterference with present-moment experiences, which is anticipated to reduce emotional reactivity, i.e., the extent to which an individual responds to a stimulus.^[28]

The Transactional Model of Stress and Coping^[29] highlights how humans employ diverse coping strategies when confronted with stressors, categorizing these strategies into three primary types: emotion-focused coping, which involves regulating emotions triggered by stress; problem-focused coping, aimed at solving the stressor; and meaning-focused coping, which cultivates positive emotions rather than suppressing negative emotions.^[30] Meaning-focused coping, in particular, triggers positive emotions that replenish mental resources, aiding in handling issues that are beyond direct control or resolution.^[31,32] Within this coping framework, mindfulness can be seen as a tool for reconsidering both the nature of the stressor and the personal resources required to overcome it.^[33]

Speculating on these theoretical points, the following hypothesis was formulated:

H1: Compared with the control group, the mindfulness training group will report significantly lower scores on the test anxiety scale at the posttest.

Mindfulness is regarded as an inherent quality linked to effective human functioning.^[34] According to the mindfulness-to-meaning theory,^[35] mindfulness practice implies decentering from internal experiences to perceive them as transient mental events rather than absolute realities. This cognitive reappraisal allows one to discover alternative meanings and perspectives that attenuate stress and increase positive affect.^[36] In this way, individuals are theorized to develop a greater capacity to respond thoughtfully to emotional turbulence and subsequently augment their psychological adaptability through activities based on mindfulness techniques. Therefore, the following hypothesis emerged:

H2: Compared with control students, students who undergo mindfulness training will report significantly higher scores on the adaptability scale at the posttest.

Research within developmental psychology and cognitive neuroscience suggests that involvement in mindfulness practices holds the promise of inducing alterations in brain structure that support academic success.^[37] A recent cross-sectional study of undergraduate students^[38] revealed that participants' decentering capacity, a magic bullet of the mindfulness concept, was attributed to a higher grade point average (GPA) as a function of lower stress. Multiple resource theory^[39] asserts that people possess a finite pool of resources for mental processes. In this context, one possible mechanism by which mindfulness training could increase learning achievement is by reducing interference while increasing the facilitation of different mental processes. For example, a mindful stance can help students concentrate on relevant information and filter out distractions, thereby curtailing the cognitive load and tweaking the encoding and retrieval of information, resulting in more effective learning.^[40,41] These considerations underpinned the hypothesis below:

H3: Compared with those in the control group, students who take part in mindfulness training exhibit increased academic performance, reflected in significantly greater GPA at the posttest.

In the mid-20th century, the cognitive interference model proposed the notion of a negative reciprocity between test anxiety and intelligence variables.^[42] Typically, anxiety engenders a heightened inclination to interpret ambiguous information in a more negative or

threatening manner. This tendency is often observed in individuals experiencing clinical depression, potentially clarifying why some individuals with symptoms of both depression and anxiety might express confusion and struggle to concentrate.^[43] Accumulating evidence shows that higher levels of test anxiety are linked to lower performance levels.^[44-46]

Students experiencing negative academic emotions, such as nervousness, sadness, anxiety, boredom, or disappointment, often focus more on perceived threats in their surroundings. This narrowed attention restricts their ability to utilize cognitive resources effectively for optimal performance in learning tasks. Consequently, these negative academic emotions may adversely affect learning results.^[47] With this in view, we propose the following hypothesis:

H4: Students' test anxiety is significantly (negatively) related to their learning performance.

In terms of personal resources, adaptability stands out as a crucial asset enabling students to navigate disruptions and achieve favorable learning outcomes.^[48] Research has associated adaptability with higher grades via increased academic engagement.^[49]

The psychology literature has proposed that achievement motives can be divided into two components: hope for success and fear of failure.^[50] On the basis of this notion, one might postulate an inverse relationship between test anxiety and adaptability because the latter can catalyze positive expectations of achieving a desired outcome while mitigating negative anticipation of failing to meet set standards, thereby enabling students to better manage their emotions in upcoming testing situations, potentially relieving test anxiety.

Empirical evidence directly linking adaptability and learning outcomes is limited, yet several studies^[51-53] suggest a positive and direct association between psychological adaptability and educational attainment. Drawing from theories of self-regulated learning, students who regulate themselves effectively are believed to adapt to various cues for optimal learning, implying that adaptability may predict better academic outcomes.^[54] On the basis of the above considerations, the present study proposes the following hypothesis:

H5: Students' adaptability will mediate the association between test anxiety and learning performance in that (a) students' test anxiety inversely predicts their adaptability and (b) adaptability positively predicts learning performance.

The theoretical framework is delineated in Figure 1.

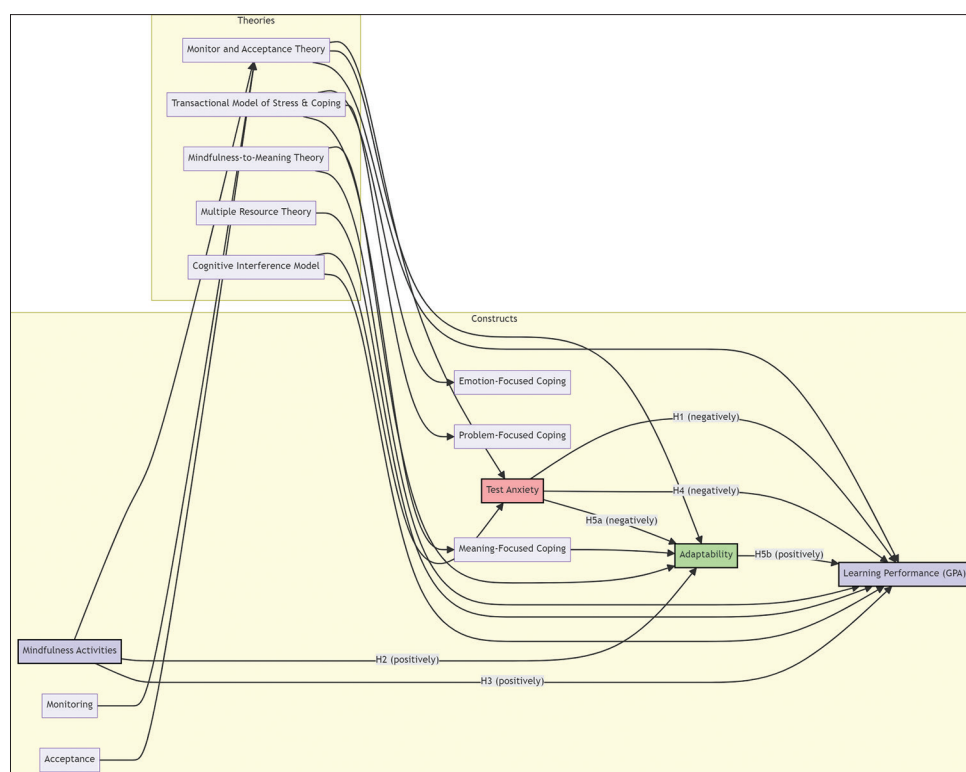


Figure 1: Hypothetical framework of the study

Materials and Methods

Study design and setting

The current study adopted a quasi-experimental design with nonrandom group assignment. Participants were recruited from the first author's university through in-person announcements and social media. Approximately 18,000 students attended the university during the 2023/2024 academic year. A required sample size of 96 was determined online, assuming a 95% confidence interval and a 10% margin of error.

Study participants and sampling

The following eligibility criteria were applied for participation in the study: (a) consent to the collection of data on academic performance; (b) completion of a paper-pencil survey at baseline and post-experiment; (c) regular participation in the intervention training (participants who missed more than one session were excluded); and (d) nonparticipation in psychological counseling or antistress practices such as yoga during the intervention period.

Between mid-September and early November 2023, a total of 483 students were approached, of whom 157 consented to participate. Among them, 149 met the inclusion criteria. Participation was voluntary and anonymous, with randomly generated participant codes linking data across time points to ensure anonymity. For

minors, consent to participate was obtained from the parent/legal guardian. Following a pretest performed through November 2023, participants with similar levels of test anxiety, adaptability, and learning outcomes were intentionally distributed across the experimental and control groups, striving to minimize disparities between them at the beginning of the study.

To avoid interrupting the experiment due to the New Year vacation, mindfulness training commenced in early February 2024 and concluded in early April 2024. As the pretest was conducted before the exam session, the posttest was also administered shortly before the final exam period, in late April 2024.

Finally, a total of 105 Russian and Kazakh-speaking students aged between 17 and 28 years ($M_{age} = 20.43$, $SD_{age} = 2.10$; 61 females, 44 males; 72 undergraduates, 33 graduates) were included in the analysis. The Chi-square test did not reveal significant between-group differences in terms of baseline characteristics. The research flow is illustrated schematically in Figure 2.

Data collection tool and technique

Test anxiety was gauged through the 16-item Multidimensional Test Anxiety Scale (MTAS),^[55] which comprises four subscales of four items each: worry, cognitive interference, tension, and physiological indicators. The participants were required to express the

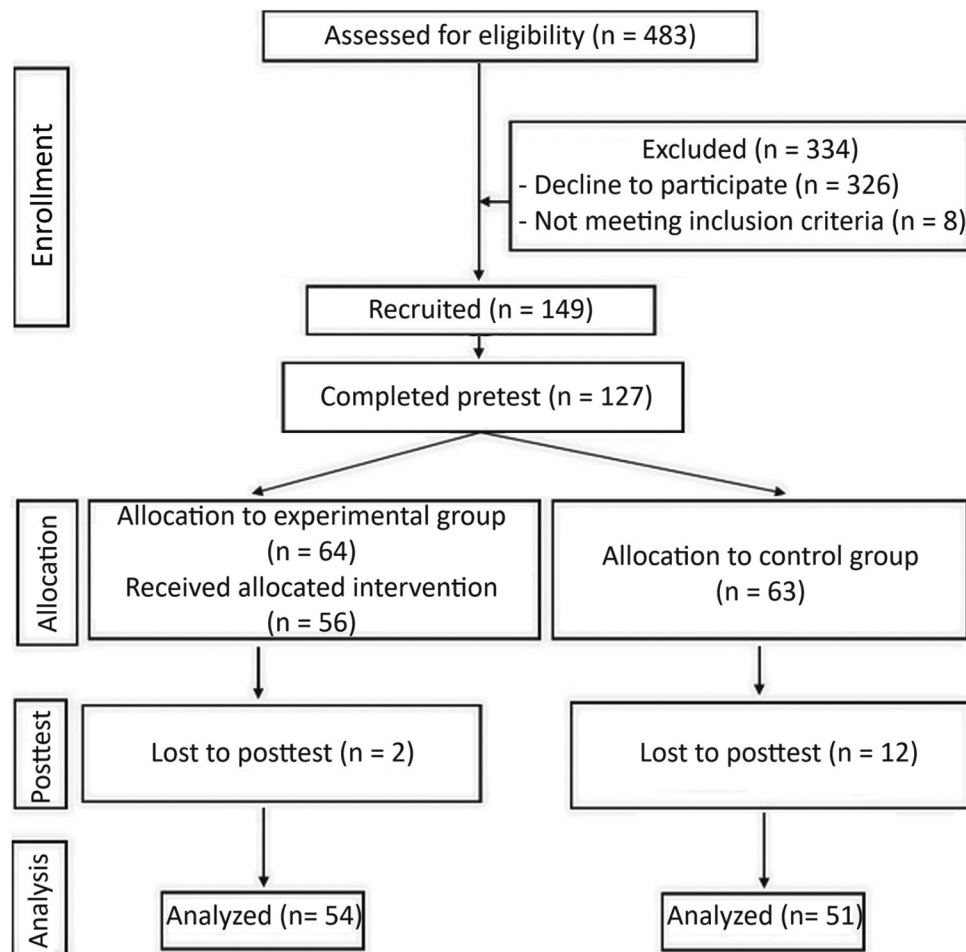


Figure 2: Flow diagram of the study

degree of their agreement with the items via a five-point response format ranging from one ("strongly disagree") to five ("strongly agree"), with higher scores implying greater anxiety in the assessment.

Adaptability was evaluated via the nine-item Adaptability Scale (AS),^[56] which focuses on cognitive-behavioral adaptability (six items) and affective adaptability (three items). The responses were recorded on a five-point scale ranging from one ("strongly disagree") to five ("strongly agree"). Higher scores are indicative of better adaptability.

Learning performance. To quantify educational outcomes, individual GPAs were obtained from the university's registration office. Researchers did not have access to any potentially identifying information.

A certified translator adapted and translated the AS and MTAS into Russian. Another translator provided a back-translation to ensure consistency with the original items. The resulting drafts were piloted with five tertiary students outside the final sample to confirm face validity. Both questionnaires demonstrated

acceptable internal consistency, with random split-half coefficients $>.70$.

Ethical consideration

The Ethics Committee of the corresponding author's institution approved the research (approval no. BH/I-ZU-31-03).

Intervention

The intervention consisted of 12 weekly 60-min sessions [the session content is outlined in Figure 3] delivered by the first author (AB) in two groups of 23–29 in a large lecture theatre over 12 weeks. The researcher holds a PhD in psychology. At the beginning of the experiment, they had three years of personal mindfulness practice and had previously led a mindfulness training group for one year. The program drew upon the traditional Buddhist meditation-based framework,^[57] incorporating concepts from Kabat-Zinn and Hanh (2009),^[58] alongside elements from holistic stress management^[59] and emotional freedom techniques.^[60] Each session combined didactic content with guided exercises facilitated by audio recordings.

Session	Topic	Content
1	Mindfulness Introduction	Mindfulness as a practice for awareness and focus in daily life. Sounds and thoughts meditation. Mindful breathing. Seated meditation.
2	Exploring Sensory Awareness	Role of senses in mindfulness. "Eating a raisin" meditation. Mindful observation: activities encouraging participants to mindfully observe everyday experiences through the senses.
3	Body-Mind Connection	Understanding connection between physical sensations, emotions, and mental states. Body scan meditation. Postural awareness practice. Being mindful in movement. Progressive muscle relaxation. Body and sound meditation.
4	Noting Practice	Introduction to RAIN (Recognize, Accept, Investigate, Not-identify) concept. Mindfulness through noting practice. Breathing space meditation.
5	Emotion Management	Understanding emotional triggers. STOP principle: distancing from instant feelings. Mindfulness of breath as a means to cope with stress. Tapping meridian points on face and hands. Breathing space meditation.
6	Positive Mindset	Internalization and concentration on positive attitudes. Self-kindness in challenging situations. Appreciative joy meditation. Loving-kindness meditation.
7	Mindful Interactions	Listening and talking in a mindful way. Listening without judgment (meditation). Sounds and thoughts meditation. Mindful walking practice.
8	Staying in the Present	Learning to ground oneself in the present moment. Seeing and hearing meditation. Walking meditation (live in the moment). Body and sound meditation.
9	Acceptance	How acceptance reduces emotional reactivity. Acceptance and change. Practice of concentrating on inner world by observing passing thoughts and identifying emotions, and how they influence each other. Seated meditation.
10	Understanding Thoughts	Thoughts as thoughts, not reality. Reinforcing RAIN and STOP practices. Body and sound meditation.
11	Sleep and Mindfulness	Accepting and overcoming insomnia. Introduction to sleep restriction and stimulus control. Mindful breathing meditation. Mindful walking practice.
12	Integration and Reflection	Reviewing the course. Integrating mindfulness into daily life. Body scan meditation. Loving-kindness meditation.

Figure 3: Overview of the mindfulness training themes and content

Data analysis

We conducted descriptive analysis and multivariate analysis of covariance (MANCOVA) on the pretest/posttest results, using pretest scores from both the experimental and control groups as covariates. Assumptions for multivariate analysis were checked, and neither Box's test for homogeneity of covariance matrices ($\chi^2 = 12.1$, $df = 6$, $P = .059$) nor the Shapiro-Wilk multivariate normality test ($W = .988$, $P = .489$) indicated violations. Additionally, no multivariate outliers were detected in the quantile-quantile plot

of the squared Mahalanobis distance and Chi-square quantiles.

The MANCOVA succeeded by separate repeated measures analyses of variance (ANOVAs) for each response variable, along with *post hoc* comparisons. Participant gender was treated as a covariate, coded as female = 1, male = 0. Partial eta squared (η_p^2) was utilized as an effect size estimate, with interpretations based on common cutoffs of .01 (small effect), .06 (moderate effect), and .14 (large effect). The significance level for ANOVA

was adjusted using a Bonferroni-corrected alpha of .016 (.05 divided by three). Both analyses were run in R.

Structural Equation Modelling (SEM) was applied to analyze the relationships among the dependent variables using the Process plug-in.^[61] Pooled pretest and posttest data from both groups were utilized to estimate regression coefficients for structural paths within the SEM framework, accounting for variations within and between the control and experimental groups. As is commonly accepted, path coefficients falling within the .10–.20 range denote a small effect, while those between .30–.40 imply a medium effect, and values above .50 indicate a large effect.^[62] The mediating effect significance was determined via the bias-corrected percentile bootstrap method with 1,000 replications. An acceptable model fit was evinced by comparative fit index values close to 1.0, root mean square error of approximation values below .08, and standardized root mean square residual values below .08.

Results

Variance statistics

Table 1 presents descriptive statistics on the outcomes under study. MANCOVA revealed a significant effect of the explanatory variable on the response variables, Pillai's trace = .363, $F(3, 98) = 18.618$, $P < .001$.

The next step was a repeated-measures ANOVA, which revealed no significant main effect of time on students' test anxiety scores ($F(1, 102) = 0.70$, $P = .405$, $\eta_p^2 = 0.007$). Overall, the interaction effect between time and group was significant ($F(1, 102) = 14.12$, $P < .001$, $\eta_p^2 = .122$), which suggests that the effect of time on the response variable differed between the groups. A post hoc test revealed that students who participated in mindfulness training significantly reduced their self-reported test anxiety from the pretest to the posttest ($t = 3.067$, $P = .017$), whereas the progress in the control group was insignificant ($t = -2.226$, $P = .169$). Additionally, there

was a significant main effect of the group on test anxiety scores ($F(1, 102) = 25.74$, $P < .001$, $\eta_p^2 = .202$). A post hoc test revealed that at the posttest, the mean test anxiety level in the mindfulness condition was significantly lower than that in the non-training conditions ($t = 6.26$, $P < 0.001$), which corroborates H1. Gender did not significantly influence the outcome ($F(1, 102) = 2.83$, $P = .096$, $\eta_p^2 = .027$). Figure 4 illustrates the temporal changes in terms of test anxiety in both groups.

The ANOVA revealed no significant main effect of time on participants' adaptability ($F(1, 102) = 2.64$, $P = .108$, $\eta_p^2 = 0.025$). The interaction effect between time and group was also not significant ($F(1, 102) = 1.97$, $P = .164$, $\eta_p^2 = 0.019$). However, the main effect of the group on this dependent variable was significant ($F(1, 102) = 10.90$, $P = .001$, $\eta_p^2 = .097$). A post hoc test revealed that after the study, the mean adaptability score in the experimental group was significantly greater than that in the control group ($t = -3.298$, $P = .007$), which supports H2. Gender did not significantly affect the measured variable ($F(1, 102) = 0.038$, $P = .845$, $\eta_p^2 = .001$). Figure 5 depicts the adaptability values observed in the two groups over the study period.

With respect to students' grades, the ANOVA did not distinguish significant effects of group ($F(1, 102) = 4.13$, $P = .050$, $\eta_p^2 = 0.039$), time ($F(1, 102) = 3.85$, $P = .052$, $\eta_p^2 = 0.036$) or their interaction ($F(1, 102) = .22$, $P = .642$, $\eta_p^2 = .002$). Given this, *post hoc* analyses were not performed. Gender also had no significant effect on GPAs ($F(1, 102) = .11$, $P = .741$, $\eta_p^2 = .001$). Thus, H3 was contradicted by the analyses. Learning performance data for the mindfulness and nontreatment conditions are shown in Figure 6.

Mediation analysis

As can be concluded from the parameter estimates in Table 2 and Figure 7, all the between-variable relationships were approximately zero. Therefore, neither H4 nor H5 were supported by the collected data.

Discussion

This study sought to explore the effects of a 12-week mindfulness training course on university students' test anxiety, adaptability, and educational success and whether test anxiety is connected to objective learning indicators directly or through adaptability. To our knowledge, mindfulness-based interventions that specifically target the triangle of learning productivity, test anxiety, and adaptability are underrepresented. In the hitherto published works on mindfulness-based interventions targeting learning-related affect, the variable of interest is generally stress, depression, or anxiety. However, the latter is prevalently trait anxiety,

Table 1: Descriptive statistics of the measured variables

Variables and Phases	Mean±SD	
	Mindfulness group (n=54)	Control group (n=51)
Test anxiety		
Pretest	3.43±0.43	3.51±0.44
Posttest	3.20±0.41	3.68±0.32
Adaptability		
Pretest	2.58±0.34	2.46±0.42
Posttest	2.71±0.56	2.42±0.49
Learning performance		
Pretest	3.73±0.33	3.80±0.25
Posttest	3.84±0.30	3.96±0.44

SD=Standard deviation

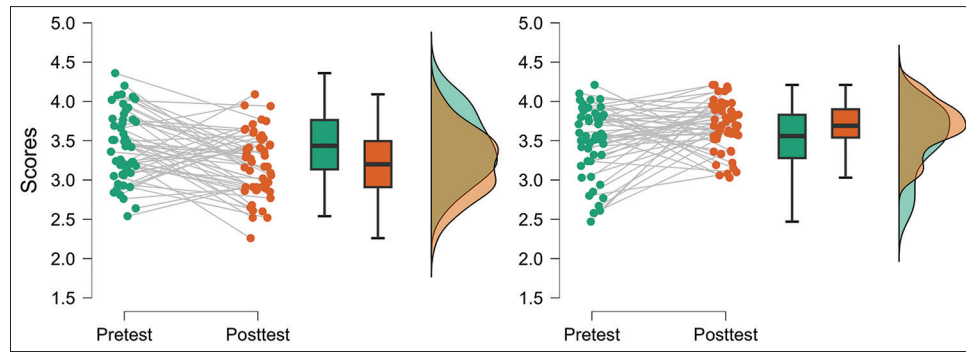


Figure 4: Test anxiety scores: mindfulness group (left side) vs comparison condition (right side)

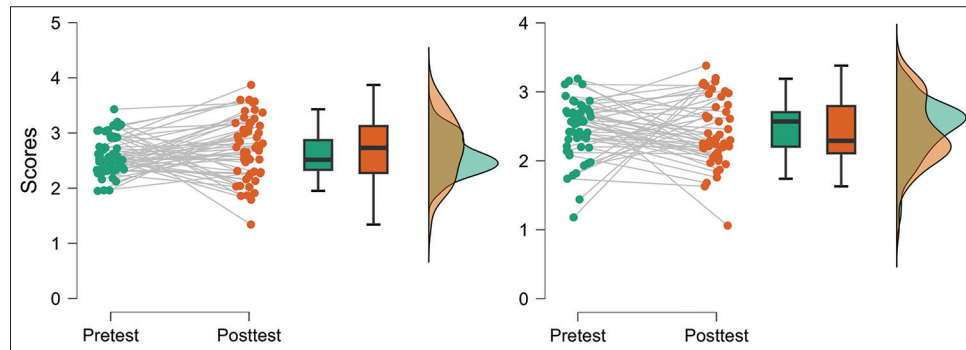


Figure 5: Adaptability scores: mindfulness group (left side) vs comparison condition (right side)

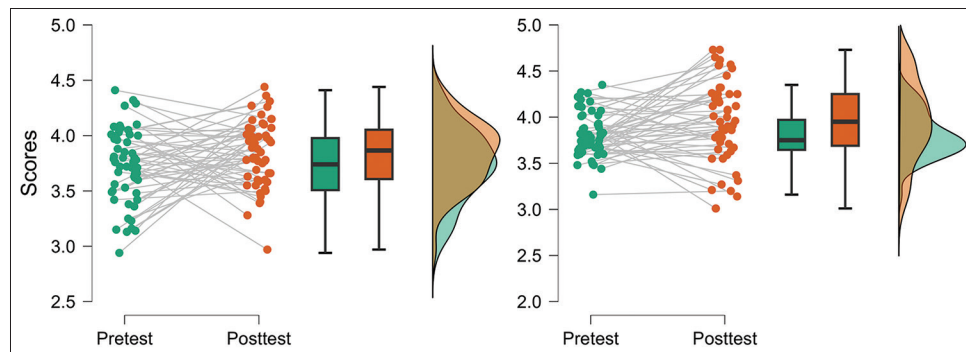


Figure 6: Grade point averages: mindfulness group (left side) vs comparison condition (right side)

Table 2: Descriptive statistics and Pearson correlations for survey responses ($n=307$)

	Estimate	Std. Error	Z	P	95% CI	
					Lower	Upper
Direct effects						
Test anxiety → Learning performance	0.004	0.054	0.079	0.937	-0.112	0.116
Indirect effects						
Test anxiety → Adaptability → Learning performance	0.008	0.008	1.011	0.312	-0.002	0.036
Total effects						
Test anxiety → Learning performance	0.012	0.055	0.224	0.823	-0.100	0.119
Path coefficients						
Adaptability → Learning performance	-0.084	0.050	-1.668	0.095	-0.190	0.014
Test anxiety → Learning performance	0.004	0.054	0.079	0.937	-0.112	0.116
Test anxiety → Adaptability	-0.095	0.074	-1.27	0.204	-0.254	0.048

CI=Confidence interval

state anxiety, or general anxiety, and for this outcome, meta-analyses report mostly low-to-moderate amplitude

effect sizes.^[63-65] Moreover, studies directly addressing the impact of mindfulness training on students' test anxiety

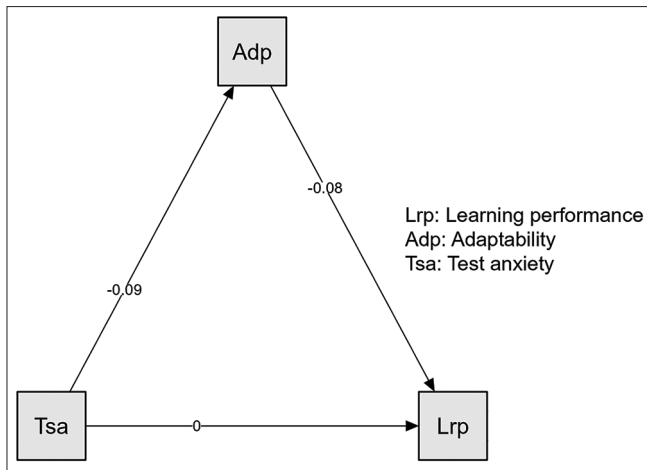


Figure 7: Mediation model estimates

are limited. For example,^[66] allocated college students to three- to six-week online mindfulness programs, which significantly lessened their test anxiety relative to the initial assessment. In a mixed-design study,^[67] college students received either offline or online mindfulness training for eight weeks, and both practices culminated in significant within-group test anxiety decline. An investigation^[68] engaged university students with test anxiety in mindfulness-based coloring, which eventually led to a substantial drop in the measured outcome. As such,^[69] eight weekly sessions of mindfulness-based stress reduction therapy were administered to female high school students, resulting in almost halving post-experimental test anxiety scores compared with those at baseline, sustaining this reduction at the three-month follow-up. Additionally,^[70] reported a significant decrease in university students' test anxiety following seven consecutive days of daily mindful breathing sessions.

In line with this existing evidence, our empirical work also favors the beneficial influence of a mindfulness program on self-reported test anxiety. This underscores the practical application of mindfulness interventions in alleviating students' test-related stress. Our work contributes further evidence to support the use of mindfulness training in higher education, assisting students in effectively managing test anxiety and potentially mitigating risky coping strategies such as substance use.^[71]

Research on the effects of mindfulness practice interventions on psychological adaptability is surprisingly rare, and we were unable to find one involving students but rather nurses^[72] and prekindergarten children,^[73] with a significant adaptability improvement in the former and no improvement in the latter. The current study lends support to the helpfulness of mindfulness practices for increasing adaptability scores. Hence, this

paper contributes to the body of knowledge highlighting the advantages of mindfulness training for students' adaptability.

Concerning educational outcomes, a recent meta-analysis of mindfulness trials^[74] revealed a medium combined effect size on GPA. Our study failed to address this point since there were no notable within-group or between-group differences throughout the experiment. A possible explanation is that although the mindfulness intervention was good for emotional regulation and adaptability, other factors that play a role in educational results, such as goal orientation, family support, or study habits, might remain unaffected. The strength of our study is that it complements the meager body of work illuminating the effects of mindfulness activities on objective indicators of learning.

The present study did not reveal significant reciprocity between test anxiety and grades. Research exploring this relationship has yielded conflicting findings. For example, Chapell *et al.*^[75] carried out a study involving 5,414 tertiary students, which revealed a small yet significant negative correlation between test anxiety and GPA. Furthermore, recent studies have revealed a flipped association between university students' anxiety and learning effectiveness.^[76,77] Nonetheless, challenging conventional wisdom, Theobald *et al.*^[78] revealed that medical students' test anxiety levels were not inversely associated with their self-reported final exam scores when the relationship was controlled for participants' objective performance on preliminary exam questions. In simpler terms, highly test-anxious individuals did not perform worse than their less anxious peers did, provided that they had a similar level of knowledge. This points to the weight of factors beyond anxiety in academic success. In support of this notion, Sung *et al.*^[79] reported a positive connection between test anxiety and achievement among low-scoring secondary school students, whereas the link was the opposite among high-scoring students, emphasizing the influence of achievement levels on the anxiety–performance interplay. In a similar vein, an observational study relying on data from the Program for International Student Assessment 2015 for Italy^[80] revealed a negative anxiety–achievement correlation, but the nuance was that high-performing students were more susceptible to the impact of test anxiety. Based on these findings, the authors argued that high test anxiety in this case might be explained not by insufficient effort or poor knowledge but rather by the idea that high-graders experience a heightened fear of failure relative to their low-performing counterparts (*ibid*). Moreover, Holzer *et al.*^[81] proposed that learners might employ active learning behavior as a coping mechanism to alleviate emotional strain due to academic pressure. Extending this to our results, namely, the lack of a contrasting

association between test anxiety and grades, we can allege that as participants in the mindfulness condition became less test anxious, their drive to overcome learning-related negative affect was dampened so that learning success efforts that—had they been made, might lead to greater learning gains—were not subsequently made. Our study therefore chips into the body of evidence challenging the negative correlation between test anxiety and academic achievement, highlighting that depletions in test anxiety do not necessarily translate to better student outcomes.

The findings of this investigation did not support the anticipated mediating role of psychological adaptability between learners' performance and test-related anxiety. It is plausible that the mindfulness training course might have independently influenced both test anxiety and adaptability rather than establishing a direct causal chain between them.

Implications

This is the first study on a mindfulness-based intervention with academic success that tests anxiety and adaptability as response variables. The study underscores mindfulness training as a conduit for promoting student well-being. By assisting in stress and anxiety management, mindfulness training has the potential to bolster overall mental health. In particular, the positive impact on adaptability underscores the broader spectrum of benefits associated with mindfulness training. Nurturing adaptability through such interventions can furnish students with vital skills, which are essential for overcoming challenges extending beyond academic realms.

Furthermore, the study intimates that mindfulness training may contribute positively to the overarching culture of educational institutions since, by imparting mindfulness practices to students, educators can diminish test anxiety and enhance psychological adaptability, fostering a conducive and inclusive learning environment. Finally, the present research underscores the need for further exploration into the underlying mechanisms governing the impact of mindfulness training on students' test anxiety, adaptability, and academic performance, aiming to gain a more comprehensive understanding of the issue.

Limitations

This study has several limitations. First, the relatively small sample size might limit the generalizability of the results. Subsequent research efforts should replicate our study using larger and more diverse samples of university students to increase the reliability and applicability of the findings. Second, we did not control for potential variables such as learning motivation or family background. Future studies in this domain

should consider including these variables as possible confounders to obtain a more nuanced understanding of the relationships observed.

Conclusion

This study adds to the current literature on the effects of mindfulness training on students' psychological and academic outcomes, and the findings oppose some previously held assumptions and research outcomes. Despite the lack of direct impact on learning performance, mindfulness training remains a valuable resource for university students struggling with test anxiety and low adaptability. This practice can help students develop the emotional regulation skills and mental flexibility necessary to cope with academic stress and navigate the challenges of higher education. These benefits align with the recommendations of the World Health Organization and the United Nations International Children's Emergency Fund to promote mental health in educational environments. To fully understand the potential benefits of mindfulness for psychological and academic outcomes, further research is necessary. This should include longitudinal studies to explore the long-term effects of mindfulness practices on academic performance and to test various causal models. Collaborative efforts between educational institutions and health policy experts are recommended to establish evidence-based guidelines for integrating mindfulness training into university curricula. Through such interdisciplinary efforts, we can better understand the mechanisms through which mindfulness training influences students' psychological well-being and academic success, ensuring that interventions are both effective and aligned with broader educational goals.

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Conflicts of interest

There are no conflicts of interest.

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